Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - II

Course Code: BCA104-2 **Max Marks: 100**

Course Name: DISCRETE MATHEMATICS **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Demonstrate a working knowledge of set notation and elementary set theory and recognize the connection between set operations and logic.

CO2: Apply the different properties of injections, surjections, bijections, compositions, and inverse functions.

CO3: Understand different types of matrices and operations among matrices.

CO4: Determine when a relation is reflexive, symmetric, antisymmetric, transitive, apply the properties of equivalence relations and partial orderings, and explain the connection between equivalence relations.

CO5: Effective express the elementary concepts and results of Number theory.

Section A **3*8=24**

1)	Show that the set of odd positive integers is a countable set.	**3**	(CO2, L2)
2) a)	Show that the set of odd positive integers is a countable set.	**3**	(CO2, L2)
	OR		
b)	Find the greatest common divisor of (12, 36, 60, 108).	**3**	(CO4, L4)
3)	Give an example of an equivalence relation on a set whose equivalence classes are finite.	**3**	(CO3, L2)
4)	How can a directed graph of a relation on a finite set be used to determine whether the relation is reflexive, symmetric?	**3**	(CO3, L2)
5) a)	How can a directed graph of a relation on a finite set be used to determine whether the relation is reflexive, symmetric?	**3**	(CO3, L2)
	OR		
b)	State division algorithm with an example for integers.	**3**	(CO4, L1)
6)	Find the symmetric difference of {, , } and {, , }.	**3**	(CO1, L1)
7) a)	Give an example of an equivalence relation on a set whose equivalence classes are finite.	**3**	(CO3, L2)
	OR		
b)	Show that the set of odd positive integers is a countable set.	**3**	(CO2, L2)
8)	Define POSET of a set with an example.	**3**	(CO2, L3)

Section B

8*7=56

9)	State Euclidean algorithm for integers, hence obtain the greatest common divisor of [414, 662].	**8**	(CO4, L4)
10) a)	Let and be relations on a set represented by the matrices 1 2 0 1 0 0 1 0 = [1 1 1] and = [0 1 1], then find the matrices that 1 2 1 0 0 1 1 1 represent a) b) ∩ c) d) e) 1 2 1 2 2 1 1 1 1 2	**8**	(CO3, L3)
	OR		
b)	In a survey of 60 people it was found that, 25 read weekly magazines, 26 read fortnightly magazines, 26 read monthly magazines, 9 read both weekly and monthly magazines,11 read both weekly and fortnightly magazines, 8 read both monthly and fortnightly magazines and 3 read all 3 magazines. Develop the Venn diagram and hence determine the number of people who read a) at least one of the three magazines. b) exactly one magazine.	**8**	(CO1, L5)
11)	Let and be relations on a set represented by the matrices 1 2 0 1 0 0 1 0 = [1 1 1] and = [0 1 1], then find the matrices that 1 2 1 0 0 1 1 1 represent a) b) ∩ c) d) e) 1 2 1 2 2 1 1 1 1 2	**8**	(CO3, L3)
12) a)	Prove that if and are countable sets, then also countable.	**8**	(CO2, L1)
	OR		
b)	Consider the relation = {(1, 2), (1,3), (1,4), (2,3), (2,4), (3, 4)} on the set {1, 2, 3, 4}. Find the reflexive, symmetric and transitive closures of .	**8**	(CO3, L4)
13)	State Euclidean algorithm for integers, hence obtain the greatest common divisor of [414, 662].	**8**	(CO4, L4)
14) a)	State Euclidean algorithm for integers, hence obtain the greatest common divisor of [414, 662].	**8**	(CO4, L4)
	OR		
b)	Let , : → be the relations on the sets = {1, 2, 3} and 1 2 = {1,2,3,4} defined by = {(1,1), (2,2), (3,3)} and 1 = {(1,1), (1,2), (1,3), (1,4)}. Then find 2, 2, 2 1 1 − 1, ∩ 2, 1 and its corresponding relation matrices. 2 1 2	**8**	(CO3, L3)
15)	Find the octal expansion of (12345) and hexadecimal expansions of 10 (177130) . 10	**8**	(CO4, L2)

Section C

10*2=20

16)	Draw the Hasse diagram for the poset (, |), where the set S consist {2, 4, 6, 9, 12, 18, 27, 36, 48, 60, 72}. Hence answer the following a) Find the maximal and minimal elements. b) Is there a greatest and least element? if so write it. c) Find all upper bounds of {2, 9} and write the least upper bound of {2, 9}, if it exists. d) Find all lower bounds of {60, 72} and write the greatest lower bound of {60, 72}, if it exists	**10**	(CO1, L3)
17) a)	Prove that the set of real numbers are uncountable set.	**10**	(CO1, L3)
	OR		

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b)	Draw the Hasse diagram for the poset (, |), where the set S consist {2, 4, 6, 9, 12, 18, 27, 36, 48, 60, 72}. Hence answer the following a) Find the maximal and minimal elements. b) Is there a greatest and least element? if so write it. c) Find all upper bounds of {2, 9} and write the least upper bound of {2, 9}, if it exists. d) Find all lower bounds of {60, 72} and write the greatest lower bound of {60, 72}, if it exists	**10**	(CO1, L3)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create